

About This Document

Short-answer questions with model answers for the BI3 technical interview, covering databases, warehousing, ETL and data modeling.

Data Engineering Interview - Short Answers

Q1. What is the difference between a data warehouse and a data lake?

Answer: A warehouse stores structured, processed, schema-on-write data for analytics; a lake stores raw data of any format (schema-on-read), cheaper and more flexible but needs processing first.

Authenticity: Community Reported

Q2. Explain OLTP versus OLAP.

Answer: OLTP handles many short transactional reads/writes with normalized schemas; OLAP handles complex analytical queries over large historical data with denormalized schemas.

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Q3. Difference between DELETE, TRUNCATE and DROP?

Answer: DELETE removes selected rows (rollback-able, WHERE-able); TRUNCATE clears all rows fast with minimal logging; DROP removes the table structure and data entirely.

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Q4. Primary key vs foreign key?

Answer: A primary key uniquely identifies each row and cannot be null; a foreign key references another table's primary key to enforce referential integrity.

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Q5. What is normalization and why use it?

Answer: Organising data into related tables (1NF/2NF/3NF) to remove redundancy and anomalies, improving integrity; analytics often denormalizes for speed.

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Q6. Batch vs stream processing?

Answer: Batch processes large data collected over a period (nightly jobs); streaming processes events continuously in near real time (Kafka + Spark Streaming).

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Q7. What is a star schema?

Answer: A central fact table of measures and foreign keys surrounded by denormalized dimension tables; simple and fast for analytical queries.

Authenticity: Community Reported

Q8. How would you optimize a slow SQL query?

Answer: Add proper indexes, select only needed columns, filter early, avoid needless joins/subqueries, partition, read the execution plan, and materialise heavy aggregations.

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Q9. What is a surrogate key?

Answer: A system-generated unique identifier (e.g. an auto-increment id) used as a primary key instead of a natural business key, keeping joins stable when business values change.

Authenticity: Research-Backed Company Style

Q10. What is data partitioning and why does it help?

Answer: Splitting a table by a key (e.g. date) so queries scan only relevant partitions, improving performance and manageability of large datasets.

Authenticity: Community Reported

BI3 Technologies

Data Engineering Interview Q&A

Applied Role(s):

Data Engineer - Trainee

Online Assessment Preparation Material

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Every question carries an authenticity label. See the legend on the next page.

Table of Contents

About This Document	1
Data Engineering Interview - Short Answers	1
Authenticity Legend & How to Read This Document	4

Authenticity Legend & How to Read This Document

Every question and problem in this document is labelled with an **authenticity classification** so you know how closely it maps to what candidates have actually reported. Labels are assigned conservatively based on research; authored practice questions are never presented as real exam questions.

Label	Meaning
Verified Previous-Year	The specific problem/pattern is documented from actual reported assessment experiences.
Frequently Repeated	The topic or pattern type is documented as recurring across multiple hiring cycles.
Community Reported	A widely shared, role-relevant pattern reported across placement communities and forums.
Research-Backed Company Style	Authored to match the researched difficulty, style and pattern - practice, not a real past question.

The correct option in every MCQ is highlighted in green and also restated below the question. Study the pattern, not just the answer.